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Classification of Magnetic Resonance Images using Bag of Features for Detecting Dementia

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Abstract

In this paper, a model is presented for classification of Dementia brain disease using magnetic resonance imaging. Bag of features (BOF) is used for extracting the features of MRI scans, which are classified using multi-class Support Vector Machine (SVM) for distinguishing the scans into three categories as demented, mildly cognitive impaired and normal controls. The Magnetic Resonance Images (MRI) used for the classification is obtained from ADNI database. The speeded up robust features are extracted using BOF approach. The experimental results showed the achievement of competitive performance in terms of classification accuracy using the proposed methodology. The BOF approach clubbed with SVM leads to an accuracy of 93%.

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1. Introduction

Dementia is a neuropsychiatric disorder that causes decline in memory mostly in the people above the age of 65 years [1]. Globally, 50 million people are estimated to be suffering from dementia and projected to be nearly triple by 2050. Each year, 9.9 million new cases are recorded worldwide, implying a new case in every 3.2 seconds [2, 3]. Neurological Magnetic Resonance Imaging (MRI) is a technique for providing a better tool for the diagnosis and tracking the progression of dementia. In this paper, we have acquired MR Images from ADNI database, for the early

detection of dementia. Bag-of-Features (BOF) is used for the feature extraction and Support Vector Machine (SVM) for the classification of the dementia, classifying MRI scans as Demented, MCI (Mild Cognitive Impairment) and CN (Control Normal). The classification Accuracy is obtained as the performance metric of the described model.

The remaining part of the paper is sorted as follows: the literature survey is presented in Section 2, Section 3 is about the proposed methodology of the work and the collection of MRI data and the framework for evaluating the classification accuracy of the proposed method. Section 4 discusses the Experimental results. Conclusions and future work are presented in Section 5. In the recent years, machine learning and deep learning have contributed a ton in the field of medical science. A brief review of the literature has been presented in this section, considering various studies for the early detection of dementia and different available datasets used in the studies.

A systematic literature review of the machine learning and microsimulation techniques for prognosing dementia has been presented by Dallora et al. [4]. Mirzaei et al. [5] reviewed various biomarkers and image acquisition techniques for diagnosing Alzheimer's disease. Ahmed et al. [6] discussed the recent advancements and future prospects of the Diagnosis of dementia using neuroimaging and machine learning techniques. The author [7] had presented a comparison of Random Forest, J48, Naïve Bayes and Multilayer perceptron and used CFSSubsetEval for feature selection. The results proved that J48 is outshining all the classifiers for detecting dementia.

Naïve Bayes, Random Forest, MLP and SMO are used in association with feature selection techniques by Bansal et al. [8] for the early detection of dementia, which showed 98.6% accuracy with CFSSubsetEval feature selection technique, considering the Oasis dataset. The accuracy obtained is high in the above study, considering neuropsychological data of 16 features from OASIS dataset. We have considered MRI Images of brain which offers a large set of features for the classification. Aruna et al. [9] proposed a feature fusion classifier using GLCM features, Gabor features, ICA and SVM as classifier achieving higher classification accuracy. Manadhar et al. [10] used ANN and KNN model for the detection of Dementia in MRI scans using segmentation and normalization of the images and resulted into 68.91% with ANN and 81.13% with KNN. The classification of histopathology images is performed using BOF approach with local features and proposed a new strategy for the separation of basic stain components semantically [11].

A new strategy [12] of histopathology images for automatic visual mining with the use of BOF is proposed by Cruz-Roa et al., which showed BOF as a decent option for the representation of histology images, which usually allows the extraction of implicit patterns for performing automatic annotation, obtaining a 90% F-score. The extension of this technique utilizing a non-negative matrix factorization is performed in [13] using a BOF approach. The piecewise feature extraction and artificial neural networks is used in [14]. Rueda et al. [15] performs automatic classification for Alzheimer's MRI scans using Bag of Features. A comparison of deep neural network and transfer learning for detecting dementia is performed by Bidani et al. [16].

Automatic Feature Selection is followed by the combination of the SVM and a pasting votes technique of ensemble SVM classifier for detecting AD at an early stage [17]. The wavelet features are extracted from the MR images, dimensionality reduction, training-test subdivision and classification is performed using SVM by Herrera et al. [18]. In [19], a Computer Aided Diagnosis system is proposed for providing an analytic method which extracts the significant features of AD consisting three stages: feature selection, extraction, and classification. The author [20] had compared multi-layer feed forward and RBF network on the basis of noise in the data. The author [21] had presented a comparison of SVM using filter based feature selection methods and without using filter based feature selection methods. However it is not clear that how well the findings will suite the other brain diseases. In [22], the author had implemented NB, C4.5, Back Propagation Network and SVM for the prediction of CDR scores and clinical diagnosis of dementia. A feasibility study [23] for the prediction of the progressing cognitive impairment to AD is conducted using machine learning techniques and observed that some more developments are required to proof the concept approach. The Voxel Based Morphometry is used for MRI images and then feature extraction techniques are applied in [24].

From the literature survey presented above, we can clearly infer that great work has been accomplished for detecting dementia at very early stage using various machine learning algorithms. However, there is still a scope of

improvement for diagnosing dementia. For improving the diagnosing, we can consider the different available datasets and can rely upon the real world data. We can increase the size of the dataset and new techniques can be implemented for better accuracy. The irrelevant and redundant features from the available dataset can also be removed by using various feature selection techniques.

2. Methodology

In the proposed work, MRI scans are used for the classification of dementia. The feature extraction from MRI scans is performed using Bag of Features technique and SVM is used for classification. The proposed model is shown in Fig. 1 which includes the mentioned steps: (1) Data Collection, (2) Pre-processing, (3) Feature extraction, and (4) Classification.

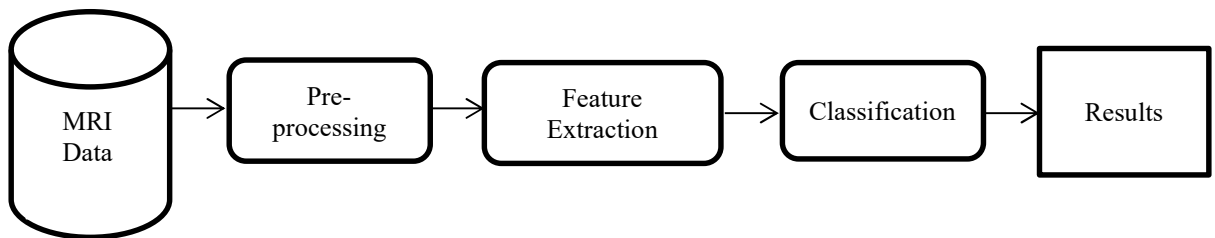


Fig. 1 Diagram of Proposed Methodology

2.1. Data Collection

The MRI scans are obtained from the ADNI [25]. MRI is a non-invasive technology which is used for producing 3-dimensional anatomical images for detecting and treating the disease. In this work, we consider the MRI images of the brain for detecting dementia. The dataset obtained is classified into 3 categories, demented, mild cognitive impairment and normal controls. A dataset of 45 images is used for the proposed model, consisting of equal number of images of each category. The example MRI scan of all the three categories is shown in Fig. 2 having left image as normal control, middle as mild cognitive impairment and right as demented.

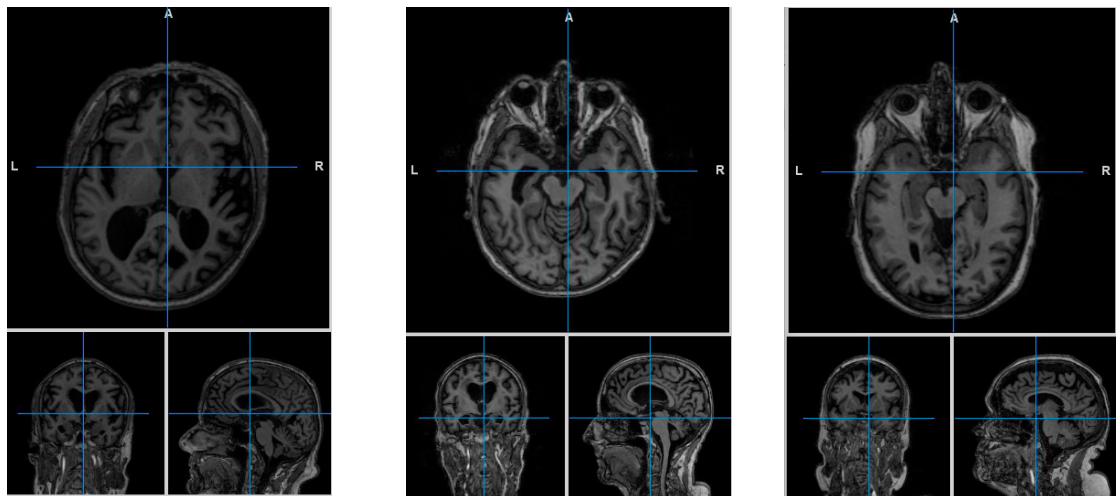


Fig. 2 Magnetic Resonance scans from ADNI. Left: normal subjects, Middle: mild cognitive impairment Right: Demented.

2.2. Image Pre-processing

The pre-processing of the images is an important factor to be considered before the feature extraction. The complexity of MRI scans is very high and it consists of a very large number of features. Therefore, we cannot use these directly for the classification and pre-processing is done before using them. The MRI images are reduced to $227 * 227 * 3$ using cubic interpolation method for making them suitable to fit in our proposed model.

2.3. Feature Extraction

Bag of Features method represents the images as an order less collections of local features [26]. It is used for feature extraction, selecting grid method for feature point locations for the extraction of speeded up robust features (SURF). A visual vocabulary is constructed for the representation of the dictionary forming clusters of features extracted from a set of training images. The feature detection and representation aspects both are consisted in SURF features. The GridStep is [8 8] and the Block Width is [32 64 96 128]. The steps for generating a BOF representation is shown in Fig. 3 which includes building vocabulary, assign terms and generate term vector. A training set is considered initially for extracting the features and these features are clustered into a visual vocabulary, where each cluster represents a term. The nearest neighbors technique is used for assigning the features to the closest terms in the vocabulary. Then the count of each term is recorded which appears in the image for creating a normalized histogram representing a term vector. Using the dataset, 103488 features are extracted using BOF approach. Bag of Features considers 80 percent of the strongest features from each category. It uses K-Means clustering for creating a 500 word visual vocabulary. A total of 500 clusters and 82791 features are extracted.

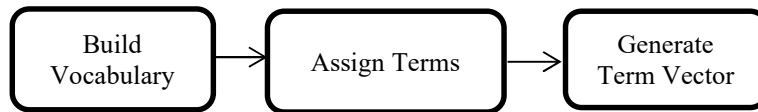


Fig. 3 Steps of Bag of Features Approach

2.4. Classification

Support Vector Machine (SVM) is the most common classifier used for the classification in the medical field. It can be used for both binary and multi-class classification [27]. It implements the idea to map the vectors nonlinearly into a very high dimensional feature space. The training is performed and on completion of training, the test data is mapped to the feature space and finally the class is then assigned [28]. The concept of decision planes for defining decision boundaries is basis of SVM. A decision plane is used for the separation of a set of objects having different class memberships. In this work, a multiclass SVM classifier is trained using an input bag with Gaussian kernel and BoxConstraint as 1.1.

3. Experimental Results

In this proposed work, MRI images from ADNI database are considered for detecting dementia and classifying into three stages: No dementia, mild cognitive impairment and dementia. There are 15 subjects in each class for the classification. The dataset is randomly distributed into training and testing datasets. The classification accuracy is used as the performance metric of the explained Learning classification model in this work which is described as follows in Eq. 1. The classification accuracy considers True Positives (T_p), True Negatives (T_n), False Positives (F_p) and False Negatives (F_n). The classification accuracy is the total number of correct predictions divided by the total number of predictions. It is one of the most commonly used metrics for the performance evaluation in the data mining.

$$\text{Accuracy} = \frac{T_p + T_n}{T_p + T_n + F_p + F_n} \quad (1)$$

The proposed model resulted an accuracy of 93%, which is overall better for the detection of dementia using BOF approach for feature extraction and SVM for classification.

Further, to demonstrate the usefulness of our proposed method for the classification of MR images as demented, mild cognitive impaired and normal subjects, we compare it with 6 state-of-the-art approaches as shown in Table 1. In 2014, Liu et al. [29] uses MRI images from ADNI dataset showing 84.40% of accuracy with single kernel SVM and 86.42% accuracy with multi kernel SVM. In [30], the author used RBM with SVM using MRI, PET and CSF ADNI data showing 91.40% accuracy. The MRI images from OASIS dataset is used in [10] in 2016 achieving an accuracy of 69.81% with artificial neural network and 81.13% with k-nearest neighbor. Both the ADNI and OASIS dataset are used in [31] showing 89.80% accuracy with SVM.

Table 1 Performance comparison of various algorithms using OASIS and ADNI data.

Approach	Year	Dataset	Classifier	Modalities	Accuracy
[21]	2014	ADNI	SK-SVM	MRI	84.40%
			MK-SVM		86.42%
[22]	2014	ADNI	RBM + SVM	MRI, PET, CSF	91.40%
[10]	2016	OASIS	ANN	MRI	69.81%
			k-NN		81.13%
[23]	2018	OASIS, ADNI	SVM	MRI	89.80%
Proposed	2019	ADNI	BOF + SVM	MRI	93%

4. Conclusion and Future Work

The proposed model uses Bag of Features for image representations for feature extraction from magnetic resonance images, in association with Support Vector Machine, for classifying the magnetic resonance imaging scan in three categories, normal controls, mild cognitive impairment and demented. The results proved the improvement of competitive performance of the method. The SURF features are extracted using bag of feature approach, using k-means clustering for creating the visual vocabulary. The Bag of Features and SVM in combination is an effective approach for speeding up the algorithm and reduce the computational complexity. Our future work will include the exploration of various datasets available for dementia and real world dataset. The SIFT, HOG as feature descriptors and other feature combinations for the analysis of discriminating visual words should be explored. The classification accuracy is used as a performance metrics in this study, whereas in the future some other performance measures can be used for evaluating the results like Kappa Statistics, ROC Curve, Precision, Recall, F-measure, etc.

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